Bruce A. Desmarais — Curriculum Vitae

Department of Political Science
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Employment

Professor, Department of Political Science and Institute for Computational and Data Sciences, Pennsylvania State University. 2020–Present.

DeGrandis-McCourtney Early Career Professor in Political Science. 2018–Present.

Research Scientist, Core Data Science, Meta Platforms (Jan-Aug 2022, on sabbatical).

Associate Professor of Political Science, Pennsylvania State University. 2015–2020.

Assistant Professor of Political Science, University of Massachusetts Amherst. 2010–2015.

Affiliations

Director, Center for Social Data Analytics, Pennsylvania State University, Fall 2022–Present

Associate Director, Center for Social Data Analytics, Pennsylvania State University, Fall 2018–Spring 2022

Director of Graduate Studies, Programs in Social Data Analytics, Pennsylvania State University, Fall 2017–Present

Faculty Affiliate, Pennsylvania State University Institute for Computational and Data Sciences, Fall 2015–Present.

Visiting Fellow, Statistical and Applied Mathematical Sciences Institute (Research Triangle Park, NC), Fall 2013.

Faculty Affiliate, UMass Amherst Computational Social Science Initiative, Fall 2010 – Spring 2015.

Associate Director, Institute for Social Science Research, University of Massachusetts Amherst, Spring 2014 – Spring 2015.

Education

Ph.D. Political Science, University of North Carolina at Chapel Hill, 2010
Specialties: Research Methods and American Politics

M.A. Political Science, University of North Carolina at Chapel Hill, 2008

B.A. Mathematical Economics and Public Policy, Eastern Connecticut State University, 2005, *Summa Cum Laude*

Publications

Book

1. Skyler J. Cranmer, Bruce A. Desmarais, and Jason W. Morgan. *Inferential Network Analysis*. Analytical Methods for Social Research. Cambridge University Press, 2020

Journal Articles

55. Yuehong Cassandra Tai, Yu-Ru Lin, and Bruce A. Desmarais. Public officials' online sharing of low-factual content: Institutional and ideological checks. *Political Communication*, Accepted

54. Sarah J. Gordon, Wayne E. Baker, Jose Uribe, Cassandra Chambers, and Bruce A. Desmarais. Networks of inclusion: Using teams and technology to create diverse social capital. *Social Networks*, Accepted

53. Issac Pollert, Johabed Olvera, Bruce A. Desmarais, Frederick J. Boehmke, and Jeffrey J. Harden. Collaborative diffusion: The dynamics of policy output in covid-19 interstate compacts. *Policy Studies Journal*, Accepted

52. Ishita Gopal and Bruce Desmarais. Legislative support for environmental policy innovation: an experimental test for diffusion through a cross-state policy network. *Applied Network Science*, 9(1):79, 2024

51. Rebekah Fyfe and Bruce Desmarais. Causal evidence for theories of contagious civil unrest. *International Studies Quarterly*, 68(4):sqae124, 2024

50. Ishita Gopal, Taegyoon Kim, Nitheesha Nakka, Frederick J. Boehmke, Jeffrey J. Harden, and Bruce A. Desmarais. The National Network of U.S. State Legislators on Twitter. *Political Science Research & Methods*, Accepted

49. Yuehong Cassandra Tai, Roan Buma, and Bruce A Desmarais. Official yet questionable: examining misinformation in us state legislators' tweets. *Journal of Information Technology & Politics*, 21(4):597–609, 2024

48. Frederick J Boehmke, Bruce A Desmarais, Abbie Eastman, Isabelle Grassel, Jeffrey J Harden, Samuel Harper, Liam Kaboli, Hyein Ko, Elisabeth Oster, and Tracee M Saunders. Sprc19: A database of state policy responses to covid-19 in the united states. *Scientific data*, 10(1):526, 2023

47. Medha Uppala and Bruce A Desmarais. Contagion, confounding, and causality: Confronting the three c's of observational political networks research. *Political Analysis*, 31(3):472–479, 2023

46. Jeffrey J Harden, Bruce A Desmarais, Mark Brockway, Frederick J Boehmke, Scott J LaCombe, Fridolin Linder, and Hanna Wallach. A diffusion network event history estimator. *The Journal of politics*, 85(2):436–452, 2023

45. Kevin L Young, Timothy Marple, James Heilman, and Bruce A Desmarais. A double-edged sword: The conditional properties of elite network ties in the financial sector. *Environment and Planning A: Economy and Space*, 55(4):997–1019, 2023

44. Sangyeon Kim, Howard Liu, and Bruce Desmarais. Spatial modeling of dyadic geopolitical interactions between moving actors. *Political Science Research and Methods*, 11(3):633–644, 2023

43. Taegyoon Kim, Nitheesha Nakka, Ishita Gopal, Bruce A Desmarais, Abigail Mancinelli, Jeffrey J Harden, Hyein Ko, and Frederick J Boehmke. Attention to the covid-19 pandemic on twitter: Partisan differences among us state legislators. *Legislative studies quarterly*, 47(4):1023–1041, 2022

42. Markus Neumann, Fridolin Linder, and Bruce Desmarais. Government websites as data: a methodological pipeline with application to the websites of municipalities in the united states. *Journal of Information Technology & Politics*, 19(4):411–422, 2022

41. Christian S Schmid, Ted Hsuan Yun Chen, and Bruce A Desmarais. Generative dynamics of supreme court citations: Analysis with a new statistical network model. *Political Analysis*, 30(4):515–534, 2022

40. John Schoeneman, Boliang Zhu, and Bruce A Desmarais. Complex dependence in foreign direct investment: network theory and empirical analysis. *Political Science Research and Methods*, 10(2):243–259, 2022

39. Fridolin Linder, Bruce Desmarais, Matthew Burgess, and Eugenia Giraudy. Text as policy: Measuring policy similarity through bill text reuse. *Policy Studies Journal*, 48(2):546–574, 2020

38. Frederick J Boehmke, Mark Brockway, Bruce A Desmarais, Jeffrey J Harden, Scott LaCombe, Fridolin Linder, and Hanna Wallach. Spid: A new database for inferring public policy innovativeness and diffusion networks. *Policy Studies Journal*, 48(2):517–545, 2020

37. Skyler J Cranmer, Bruce A Desmarais, and Benjamin W Campbell. The contagion of democracy through international networks. *Social Networks*, 61:87–98, 2020

36. Xi Liu, Clio Andris, and Bruce A Desmarais. Migration and political polarization in the us: An analysis of the county-level migration network. *PloS one*, 14(11), 2019

35. Shana Scogin, Sarah Petersen, Jeffrey Harden, and Bruce Desmarais. modelltest: An r package for unbiased model comparison using cross validation. *Journal of Open Source Software*, 4(41):1542, 9 2019

34. Bruce A. Desmarais. Punctuated equilibrium or incrementalism in policymaking: What we can and cannot learn from the distribution of policy changes. *Research & Politics*, 6(3), 2019

33. Paul E Stillman, James D Wilson, Matthew J Denny, Bruce A Desmarais, Skyler J Cranmer, and Zhong-Lin Lu. A consistent organizational structure across multiple functional subnetworks of the human brain. *NeuroImage*, 197:24–36, 2019

32. Carisa Bergner, Bruce A Desmarais, and John Hird. Speaking truth in power: Scientific evidence as motivation for policy activism. *Journal of Behavioral Public Administration*, 2(1), 2019

31. Sayali Phadke and Bruce A Desmarais. Considering network effects in the design and analysis of field experiments on state legislatures. *State Politics & Policy Quarterly*, 19(4):451–473, 2019

30. Mia Costa, Bruce A Desmarais, and John A Hird. Public comments' influence on science use in us rulemaking: The case of epa's national emission standards. *The American Review of Public Administration*, 49(1):36–50, 2019

29. Shankar Bhamidi, Suman Chakraborty, Skyler Cranmer, and Bruce Desmarais. Weighted exponential random graph models: Scope and large network limits. *Journal of Statistical Physics*, 173(3-4):704–735, 2018

28. Bruce A Desmarais. Discussion of "inferring social structure from continuous-time interaction data". *Applied Stochastic Models in Business and Industry*, 34(2):107–109, 2018

27. Jake Bowers, Bruce A. Desmarais, Mark Frederickson, Nahomi Ichino, Hsuan-Wei Lee, and Simi Wang. Models, methods and network topology: Experimental design for the study of interference. *Social Networks*, 54:196 – 208, 2018

26. Paul E Stillman, James D Wilson, Matthew J Denny, Bruce A Desmarais, Shankar Bhamidi, Skyler J Cranmer, and Zhong-Lin Lu. Statistical modeling of the default mode brain network reveals a segregated highway structure. *Scientific reports*, 7(1):11694, 2017

25. Philip Leifeld, Skyler Cranmer, and Bruce Desmarais. Temporal exponential random graph models with btergm: Estimation and bootstrap confidence intervals. *Journal of Statistical Software, Articles*, 83(6):1–36, 2018

24. Skyler J Cranmer and Bruce A Desmarais. What can we learn from predictive modeling? *Political Analysis*, 25(2):145–166, 2017

23. James D. Wilson, Matthew J. Denny, Shankar Bhamidi, Skyler J. Cranmer, and Bruce A. Desmarais. Stochastic Weighted Graphs: Flexible Model Specification and Simulation. *Social Networks*, 49:37–47, 2017

22. Frederick J Boehmke, Abigail Matthews Rury, Bruce A Desmarais, and Jeffrey J Harden. The seeds of policy change: Leveraging diffusion to disseminate policy innovations. *Journal of health politics, policy and law*, 42(2):285–307, 2017

21. James ben Aaron, Matthew Denny, Bruce Desmarais, and Hanna Wallach. Transparency by conformity: A field experiment evaluating openness in local governments. *Public Administration Review*, 77(1):68–77, 2017

20. Skyler J Cranmer and Bruce A Desmarais. A critique of dyadic design. *International Studies Quarterly*, 60(2):355–362, 2016

19. Mia Costa, Bruce A Desmarais, and John A Hird. Science use in regulatory impact analysis: The effects of political attention and controversy. *Review of Policy Research*, 33(3):251–269, 2016

18. Bruce A. Desmarais, Jeffrey J. Harden, and Frederick J. Boehmke. Persistent Policy Pathways: Inferring Diffusion Networks in the American States. *American Political Science Review*, 109(2):392–406, 2015

17. Bruce A. Desmarais, Vincent G. Moscardelli, Brian F. Schaffner, and Michael S. Kowal. Measuring Legislative Collaboration: The Senate Press Events Network. *Social Networks*, 40:43–54, 2015
16. Bruce A. Desmarais, Raymond J. La Raja, and Michael S. Kowal. The Fates of Challengers in US House Elections: The Role of Extended Party Networks in Supporting Candidates and Shaping Electoral Outcomes. *American Journal of Political Science*, 59(1):194–211, 2015
15. Bruce A. Desmarais and John A. Hird. Public Policy's Bibliography: The Use of Research in U.S. Regulatory Impact Analyses. *Regulation & Governance*, 8(4):497–510, 2014
14. Bruce A. Desmarais and Jeffrey J. Harden. An Unbiased Model Comparison Test Using Cross-Validation. *Quality & Quantity*, 48(4):2155–2173, 2014
13. Bruce A. Desmarais and Jeffrey J. Harden. Testing for Zero-Inflation in Count Models: Bias Correction for the Vuong Test. *The Stata Journal*, 13(4):810–835, 2013
12. Skyler J. Cranmer, Tobias Heinrich, and Bruce A. Desmarais. Reciprocity and the Structural Determinants of the International Sanctions Network. *Social Networks*, 36(January):5–22, 2014
11. Bruce A. Desmarais and Skyler J. Cranmer. Micro-level interpretation of exponential random graph models with application to estuary networks. *Policy Studies Journal*, 40(3):402–434, 2012.
10. Skyler J. Cranmer, Bruce A. Desmarais, and Justin H. Kirkland. Toward a Network Theory of Alliance Formation. *International Interactions*, 38(3):295–324, 2012
9. Stuart M. Benjamin and Bruce A. Desmarais. Standing the Test of Time; The Breadth of Majority Coalitions and the Fate of U.S. Supreme Court Precedents. *Journal of Legal Analysis*, 4(2):445–469, 2012.
8. Skyler J. Cranmer, Bruce A. Desmarais, and Elizabeth Menninga. Complex Dependencies in the Alliance Network. *Conflict Management and Peace Science*, 29(3):279–313, 2012.
7. Bruce A. Desmarais and Jeffrey J. Harden. Comparing partial likelihood and robust estimation methods for the cox regression model. *Political Analysis*, 20(1):113–135, 2012.
6. Bruce A. Desmarais and Skyler J. Cranmer. Statistical inference for valued-edge networks: The generalized exponential random graph model. *PLoS ONE*, 7(1):e30136, 01 2012.
5. Bruce A. Desmarais. Lessons in disguise: Multivariate predictive mistakes in collective choice models. *Public Choice*, 151(3):719–737, 2012.
4. Bruce A. Desmarais and Skyler J. Cranmer. Statistical Mechanics of Networks: Estimation and Uncertainty. *Physica A*, 391(4):1865–1876, 2012.
3. Skyler J. Cranmer and Bruce A. Desmarais. Inferential Network Analysis with Exponential Random Graph Models. *Political Analysis*, 19(1):66–86, 2011.
2. Jeffrey J. Harden and Bruce A. Desmarais. Linear Models with Outliers: Choosing Between Conditional-Mean and Conditional-Median Methods. *State Politics & Policy Quarterly*, 11(4):371–389, 2011
1. Allison T. Freeman and Bruce A. Desmarais. Portfolio Adjustment to Home Equity Accumulation among CRA Borrowers. *Journal of Housing Research*, 20(2):141–160, 2011

Peer Refereed Proceedings

10. Yuehong Cassandra Tai, Khushi Navin Patni, Nicholas Daniel Hemauer, Bruce Desmarais, and Yu-Ru Lin. Genai vs. human fact-checkers: Accurate ratings, flawed rationales. In *Companion Publication of the 17th ACM Web Science Conference*, Accepted
9. Ahana Biswas, Yu-Ru Lin, Yuehong Cassandra Tai, and Bruce A Desmarais. Political elites in the attention economy: Visibility over civility and credibility? In *19th International Conference on Web and Social Media, ICWSM 2025*. AAAI Press, Accepted
8. Fangcao Xu, Bruce Desmarais, and Donna Peuquet. Stand: A spatio-temporal algorithm for network diffusion simulation. In *Proceedings of the 3rd ACM SIGSPATIAL International Workshop on GeoSpatial Simulation*, GeoSim '20, page 20–29, New York, NY, USA, 2020. Association for Computing Machinery

7. Sangyeon Kim, Omer F Yalcin, Samuel E Bestvater, Kevin Munger, Burt L Monroe, and Bruce A Desmarais. The effects of an informational intervention on attention to anti-vaccination content on youtube. In *Proceedings of the International AAAI Conference on Web and Social Media*, volume 14, pages 949–953, 2020

6. Bruce A Desmarais and John A Hird. Policy-relevant science: The depth and breadth of support networks. In *Complex Networks XI*, pages 385–392. Springer, 2020

5. T. Marple, B. Desmarais, and K. L. Young. Collapsing corporate confusion: Leveraging network structures for effective entity resolution in relational corporate data. In *2017 IEEE International Conference on Big Data (Big Data)*, pages 2637–2643, Dec 2017

4. C. S. Schmid and B. A. Desmarais. Exponential random graph models with big networks: Maximum pseudolikelihood estimation and the parametric bootstrap. In *2017 IEEE International Conference on Big Data (Big Data)*, pages 116–121, Dec 2017

3. Peter Krafft, Juston Moore, Bruce Desmarais, and Hanna Wallach. Topic-Partitioned Multinetwork Embeddings. In *Proceedings of the 26th Annual Conference on Neural Information Processing Systems*, 2012

2. Bruce A. Desmarais and Skyler J. Cranmer. Forecasting the Locational Dynamics of Transnational Terrorism: A Network Analytic Approach. In *Proceedings of the European Intelligence and Security Informatics Conference (EISIC) 2011*. IEEE Computer Society, 2011.

1. Bruce A. Desmarais and Skyler J. Cranmer. Consistent Confidence Intervals for Maximum Pseudolikelihood Estimators. *Neural Information Processing Systems 2010 Workshop on Computational Social Science and the Wisdom of Crowds*, 2010.

Chapters in Edited Volumes

3. Ishita Gopal and Bruce Desmarais. The network science of public policy diffusion. In Dino Christenson Janet Box-Steffensmeier and Valeria Sinclair-Chapman, editors, *The Oxford Handbook of Methodological Pluralism*. Oxford University Press, Forthcoming

2. John Schoeneman and Bruce Desmarais. Network modeling: estimation, inference, comparison, and selection. In Luigi Curini & Robert Franzese, editor, *The SAGE handbook of research methods in political science and international relations*, chapter 46, pages 876–894. SAGE Publications, London, 2020

1. Bruce A. Desmarais and Skyler J. Cranmer. Statistical inference in political networks research. In Jennifer Nicoll Victor, Alexander H. Montgomery, and Mark Lubell, editors, *The Oxford Handbook of Political Networks*. Oxford University Press, 2017

Software

4. Shana Scogin, Sarah Petersen, Jeff Harden, and Bruce A. Desmarais. *modeLLtest: Compare Models with Cross-Validated Log-Likelihood*, 2020. R package version 1.0.3

3. Fridolin Linder and Bruce A. Desmarais. *NetworkInference: Inferring latent diffusion networks*, 2017. R package version 1.1.1

2. Matthew J. Denny, James D. Wilson, Skyler Cranmer, Bruce A. Desmarais, and Shankar Bhamidi. *GERGM: Estimation and Fit Diagnostics for Generalized Exponential Random Graph Models*, 2016. R package version 0.10.5

1. Philip Leifeld, Skyler J. Cranmer, and Bruce A. Desmarais. *xergm: Extensions for Exponential Random Graph Models*, 2016. R package version 1.7.0

Presentations

Invited Talks

23. "Public Officials' Online Sharing of Low-factual Content: Institutional and Ideological Checks" Department of Political Science, Series on AI-Assisted Data Analysis for the Study of Politics and Government, The Ohio State University, April 14th, 2025.
22. "Public Officials' Online Sharing of Low-factual Content: Institutional and Ideological Checks" School of Data Science, University of North Carolina at Charlotte, January 23rd, 2025.
21. "Digitally Accountable Public Representation, Findings and takeaways for election season." School of Public Affairs, Penn State Harrisburg, September 25th, 2024.
20. "Network Effects Panel Regression." Department of Political Science, Vanderbilt University, September 13th, 2024.
19. "Causal Evidence for Theories of Contagious Civil Unrest." Summer Institutes in Computational Social Science, University of Rochester. May 18th, 2024.
18. "Network Effects Panel Regression." Center for Political Studies, University of Michigan, November 8th, 2023.
17. "Network Effects Panel Regression." Stochastic Modeling and Computational Statistics Speaker Series, Pennsylvania State University Department of Statistics, November 3rd, 2023.
16. Workshop roundtable participant, "Lessons Learned from UNC," American Politics Research Group 25th Anniversary Celebration, UNC Chapel Hill, September 22nd, 2023.
15. "Network Event History Analysis with application to Public Policy Diffusion." Statistical Network Science with Applications conference, Pennsylvania State University. May 19th, 2023.
14. "Predicting Dyadic and Geopolitical Interactions Between Spatially Moving Objects." Boğaziçi University (virtually), November 17th, 2021.
13. "Attention to the COVID-19 pandemic on Twitter: Partisan differences among U.S. state legislators." Facebook Core Data Science, March 25th, 2021.
12. "Network Event History Analysis for Modeling Public Policy Adoption with Latent Diffusion Networks." Quantitative Social Science Colloquium , Princeton University, April 12th, 2019.
11. "Complex Dependence in Foreign Direct Investment: Network Theory and Empirical Analysis." Workshop on Modeling Spatial and Network Interdependence in International Relations, University of St. Gallen, December 8, 2018.
10. "Complex Dependence in Foreign Direct Investment: Network Theory and Empirical Analysis." ISA Workshop on Modeling Spatial and Network Interdependence in International Relations, San Francisco, April 3, 2018.
9. "A Network Model for Textual Communications Application to Government Email Corpora." Center for Statistics and the Social Sciences, University of Washington, Seattle. April 19, 2017.
8. "Rountable on Federalism, Intergovernmental Affairs, and Public Administration." Midwest Political Science Association Annual Conference. April 8, 2016.
7. "Campaign Finance and Primary Elections." Meeting of the Campaign Finance Task Force, Stanford University. February, 4, 2016.
6. "Entity Ambiguity." Big Data and Large Corporate Networks: Enhancing Data Quality Through Standards for the Research Community, University of Amsterdam, Amsterdam. December 15, 2015.
5. "Communication Network Content and Structure: A Modeling Approach with Application to Gender Mixing in Local Government Internal E-Mail Communication." Northeast Political Methodology Meeting, New York University, May 2, 2014.
4. "Reflections on the Predictive Modeling of Interstate Conflict." ISA Working Group on Forecasting International Events, Toronto, March 25, 2014.

3. "Inferring Policy Diffusion Networks in the American States." New Frontiers in Policy Diffusion Conference, University of Iowa, March 14-15, 2014.
2. "Topic-Specific Latent Space Modeling of Government Communication Networks." Department of Computational Social Science, George Mason University, January 15, 2014.
1. "Inferring Policy Diffusion Networks in the American States." Conference on Causality in Political Networks, University of Chicago, May, 10 2013.

Conference Presentations

9. "Network Effects Panel Regression: A New Method Applied to Political Economy." 2024 American Political Science Association Annual Meeting
8. "Identification and analysis of anti-globalization rhetoric by U.S. House Members." Annual Meeting of the Society for Political Methodology, 2023, Stanford University.
7. "The National Network of U.S. State Legislators on Twitter," European Social Networks Conference, University of Greenwich, London. September 2022.
6. "The Network Science of Public Policy Diffusion." American Political Science Association Annual Meeting, Virtual, October 2021.
5. "The Diffusion Network Event History Model for the Study of Policy Adoption." American Political Science Association Annual Meeting, Washington, DC, September 2019.
4. "Diffusion of Candidates in Campaign Finance Networks." Political Networks Conference, 2018, George Mason University.
3. "A Network Model for Textual Communications Application to Government Email Corpora." Annual Meeting of the Society for Political Methodology, 2017, University of Wisconsin.
2. "Models, Methods and Network Topology: Experimental Design for the Study of Interference." Southern Political Science Association Annual Meeting. San Juan, PR. January 7, 2016.
1. "Modeling Interpersonal Government Communication Networks." Southern Political Science Association Annual Meeting. New Orleans, LA. January, 2015.

Invited Workshops

13. Network Analysis session in the MIDAS-ICPSR Social Data Science Summer Academy (August 2024)
12. Network Analysis in Human Subjects Research (ICPSR Summer Program, 2024 & 2025)
11. Machine Learning (Statistical Horizons, (December 2023, June 2024, January 2025, June 2025)
10. Recent Advances in Network Analysis (ICPSR Summer Program, 2023)
9. Introduction to Network Analysis (Oxfam, April 2023)
8. Introduction to Network Analysis in R (Political Network Conference, 2016–2019)
7. Network Analysis: Statistical Inference with Exponential Random Graph Models (UNC Chapel Hill, Odum Institute, May 2014)
6. TERGM: Exponential Random Graph Models for Dynamic Network Data (George Mason University, January 2014)
5. Advanced Network Analysis (ICPSR Summer Program, 2012–2015, 2018)
4. Introduction to Network Analysis (UNC Chapel Hill, Odum Institute, November 2013)
3. TERGM: Exponential Random Graph Models for Dynamic Network Data (Political Networks Conference, June 2013)
2. Advanced Bayesian Statistics (first two of four weeks) (ICPSR Summer Program, 2011)
1. Network Analysis Workshop, (The Ohio State University, May 2011)

Grants

15. NSF Award Number SES-2449569. "Collaborative Research: Belief Systems Governing Climate Adaptation Behaviors: Inference, Analysis, and Intervention," effective 06/01/2025; $148,477.
14. NSF Award Number SES-2318460. "Collaborative Research: HNDS-I: Digitally Accountable Public Representation," effective 09/01/2023; $649,380. PI with Sarah Rajtmajer and Kevin Munger, Co-PIs.
13. NSF Award Number SES- 2148215. "Collaborative Research: Patterns, Context, and Secondary Impacts of State Policy Responses to the Pandemic," effective 07/01/2022; $366,181.
12. NSF Award Number SES- 2028675. "RAPID: Collaborative Research: The Diffusion of State Policy Responses to the 2019 Novel Coronavirus," effective 05/15/2020; $16,417.
11. Institute for CyberScience Seed Grant, Pennsylvania State University,"Measuring Scalable Social Connectivity in Complex Networks of Intercity Flows," effective June 30, 2017 and expires June 30, 2018. Co-PI, Clio Andris. $24,537.
10. NSF Award Number SES-1637089. "RIDIR: Collaborative Research: DAPPR: Diffusion Analytics for Public Policy Research" effective October 1, 2016 and expires September 30, 2019. Penn State total: $142,425.
9. NSF Award Number SES-1558661. "Collaborative Research: An Expanded Framework for Inferring Public Policy Diffusion Networks." Effective June 15, 2016 and expires May 31, 2018. Penn State total: $178,953.
8. Hewlett Foundation. "Task Force Funds." effective October 15, 2015 and expires October 14, 2016. Penn State total: $77,658.
7. NSF Award Number SES-1357606. "Collaborative Research: Specification and Estimation of Exponential Family Random Graph Models for Weighted Networks." effective April 15, 2014 and expires April 14, 2015. UMass Amherst total: $29,983.
6. Russell Sage Foundation. "The Revolving Door in Financial Regulation: Elite Networks and the Consequences of Unequal Access on Policymaking." effective January 01, 2015 and expires December 31, 2015. UMass Total: $77,658.
5. NSF Award Number CISE-1320219. "Organizational Responsiveness to Open Outside Input: A Modeling Approach based on Statistical Text and Network Analysis." effective September 1, 2013 and expires August 31, 2016. Co-PI with Hanna Wallach (UMass Amherst; PI). Total: $499,326.
4. NSF Award Number SES-1360104. "Scientific Evidence in Regulation and Governance." effective May 1, 2014 and expires May 1, 2017. PI with John Hird (UMass Amherst; Co-PI). Total: $527,233.
3. Faculty Research Grant, University of Massachusetts Amherst Office of Research Development, 01/2013–12/2013. Total: $14,560
2. Proposal Preparation Grant, University of Massachusetts Amherst College of Social and Behavioral Sciences, 01/2011–12/2011. Total: $10,000
1. Research Support Grant, University of Massachusets Amherst College of Social and Behavioral Sciences, 01/2011 – 05/2011. Total: $4,000

Awards

5. 2025 Raymond Lombra Award for Distinction in the Social or Life Sciences, College of Liberal Arts, Pennsylvania State University
4. 2023 Best Book Award, honorable mention, Political Networks Section of the American Political Science Association. Awarded in recognition of *Inferential Network Analysis*, Cambridge University Press, 2021 (co-authored with Skyler Cranmer and Jason Morgan).

3. 2015 Jack Walker Award. Recognizes an article published in the last two calendar years that makes an outstanding contribution to research and scholarship on political organizations and parties. Awarded by the Political Organizations and Parties Section of the American Political Science Association. Co-winner with Michael Kowal and Ray La Raja for, "The Fates of Challengers in US House Elections: The Role of Extended Party Networks in Supporting Candidates and Shaping Electoral Outcomes." Published in the *American Journal of Political Science*, 2015.
2. Best Conference Paper on Political Networks, 2013. Awarded by the Political Networks Section of the American Political Science Association. Co-winner with Jeffrey J. Harden and Frederick J. Boehmke for, "Inferring Policy Diffusion Networks in the American States." Presented at the Annual Meeting of the Society for Political Methodology, University of Virginia, 2013.
1. Fellow for the Program on Computational Methods for the Social Sciences, Statistical and Applied Mathematical Sciences Institute, Fall 2013.

Patents

1. United States Patent; US 11,062,450 B2; Jul.13, 2021. SYSTEMS AND METHODS FOR MODELING NEURAL ARCHITECTURE. Skyler Cranmer, Bruce Desmarais, Shankar Bhamidi, James Wilson, Matthew Denny, Zhong - Lin Lu, Paul Stillman.

Courses Taught

9. Machine Learning for Political Science (Graduate: 2021, 2023)
8. Approaches and Issues in Social Data Analytics (Graduate: 2022, 2023, 2024, 2025)
7. Research Design for Social Data Analytics (Undergraduate: 2016, 2018)
6. Statistical Methods for Political Research (Graduate: 2018, 2019, 2020, 2021)
5. Political Network Analysis (Graduate: 2013, 2014, 2016, 2017, 2020, 2022, 2024)
4. Big Social Data: Approaches and Issues (Graduate: 2015, 2016, 2017)
3. Congress and the Legislative Process (Undergraduate: 2012, 2013, 2014)
2. Introduction to Quantitative Analysis (Graduate: 2012, 2014)
1. Official Secrecy in the United States, (Undergraduate: 2010, 2012)

Students Advised

Undergraduate honors thesis advisees

2. Motunrayo Bamgbose-Martins, Penn State, 2020.
1. Sascha Alach, UMass Amherst, 2014.

Dissertation advisees

10. Nicholas Hemauer, Penn State, PhD expected 2027.
9. Nitheesha Nakka, Penn State, PhD expected 2025.
8. Ishita Gopal, Penn State, PhD awarded 2023. Current appointment: Postdoctoral Research Associate, Transdisciplinary Institute in Applied Data Sciences, Washington University in St. Louis
7. Sangyeon Kim, Penn State, PhD awarded 2023. Curent appointment: Postdoctoral Researcher, Observatory on Social Media, Indiana University.
6. Taegyoon Kim, Penn State, PhD awarded 2022. Current appointment: Assistant Professor, School of Digital Humanities and Computational Social Sciences, KAIST
5. Omer Yalcin, Penn State, PhD awarded 2021. Current appointment: Faculty Fellow, Data Analytics and Computational Social Science, University of Massachusetts Amherst
4. Ted Hsuan Yun Chen, Penn State, PhD awarded 2019. Current appointment: Assistant Professor, Department of Environmental Science and Policy, George Mason University

3. Matthew Denny, Penn State, PhD awarded 2019. Current appointment: Staff Research Scientist, Meta
2. Fridolin Linder, Penn State, PhD awarded 2018. Current appointment: Research Scientist/Software Engineer, Meta
1. Zachary Jones, Penn State, PhD awarded 2017. Current appointment: Senior Data Scientist, Amazon.

Postdoctoral Scholars Supervised

4. Isaac Pollert, 2022-2024. Current appointment: Assistant Teaching Professor of Political Science, Pennsylvania State University
3. Yuehong Cassandra Tai, 2022-2024. Current appointment: Research Assistant Professor, Center for Social Data Analytics, Pennsylvania State University
2. Medha Uppala, 2019-2022. Current appointment: Senior Statistician, Westat.
1. Howard Liu, 2019-2020. Current appointment: Assistant Professor, Department of Political Science, University of South Carolina.

External Service

22. Program reviewer, Artificial Intelligence & Policy Analysis program, University at Buffalo
21. Mentor for the The Society for Political Methodology's Expansions Initiative, Fall 2023 & Fall 2024
20. Program reviewer, Data Science program at George Washington University, March 2023
19. Program reviewer, Data Analytics program, Ohio Wesleyan University, June 2023
18. Mentor for the The Society for Political Methodology's Expansions Initiative, Fall 2022
17. Emerging Scholar Award Committee Member, State Politics & Policy section of the American Political Science Association, 2022
16. Host, Annual Conference on Political Networks (PolNet) 2021
15. Editorial Board Member, *State Politics & Policy Quarterly*, 2020 – Present.
14. 2019–2020 Member, Christopher Z. Mooney (state politics) Dissertation Prize Selection Committee
13. 2018–2019 President, Political Networks section of the American Political Science Association.
12. Ninth Annual Conference on New Directions in Analyzing Text as Data, 2018, Program Committee Member.
11. 2018 Best Graduate Student Poster Award Committee Chair, State Politics & Policy Conference.
10. 2017 Best Paper Award Committee Chair, Political Organizations and Parties section of the American political Science Association.
9. Program Committee Member, European Symposium on Computational Social Science, 2017
8. Program Committee Member, International Conference on Social Informatics, 2014, 2016, 2017
7. Editorial Board Member, *State Politics & Policy Quarterly*, 2011 – 2014.
6. Political Networks Conference Fellowship Committee, member 2012, chair 2013.
5. Political Networks Conference Program Committee Co-chair, 2014.
4. Program Committee Member, Workshop on Computational Social Science and the Wisdom of Crowds. Held at Neural Information Processing Systems 2011, Sierra Nevada, Spain.
3. National Science Foundation Review Panel Member, Spring 2013, Summer 2014, Spring 2021, Spring 2023.

2. Reviewer: *American Political Science Review, Science, American Journal of Political Science, Political Analysis, Journal of Politics, British Journal of Political Science, Conflict Management and Peace Science International Organization, International Studies Quarterly, Proceedings of the National Academy of Science, Science, Public Administration Review, Social Networks, State Politics & Policy Quarterly, PLoS ONE, Policy Studies Journal, Political Research Quarterly, American Politics Research, Public Choice, Journal of Computational and Graphical Statistics, Journal of Machine Learning Research, Journal of Statistical Software, The Stata Journal, Journal of Public Policy*, National Science Foundation, Swiss National Science Foundation, National Science Center (Poland).

1. Member: American Political Science Association (sections on Political Methodology and Political Networks) and the Midwestern Political Science Association